

interaction. These charged particles may collide with electrons, which remove them from their parent atoms, or transfer energy to an electron by interaction of electric fields.

How do you choose a detection device?

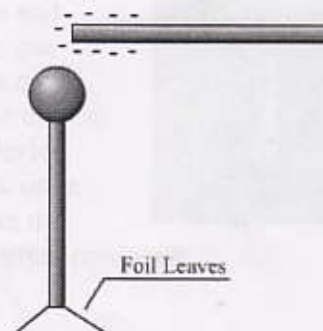
Important considerations for choosing a particular type of detection device include the application, the type of radiation, the energy of the radiation, and the level of sensitivity needed. Remember from previous discussion that radiation exists as waveforms with varying energies and may be either particulate or electromagnetic in nature.

There are three types of radiation detection devices:

- o The Electroscope
- o The Cloud Chamber
- o Other Detection Devices

THE ELECTROSCOPE

Marie Curie used an electroscope to study the radioactivity of Uranium ores. The electroscope is a fairly simple device comprised of a metal rod with two thin leaves attached to one end. If the electroscope is given a negative charge, the metal leaves will separate from each other. It is this characteristic that makes the electroscope useful as a detection device. A negatively charged electroscope will discharge when ions in the air remove electrons from it, and consequently, a positively charged electroscope will discharge when it takes electrons from the air around it. The rate of discharge of the electroscope is a measure of ions in the air and can be used as a basis of measurement and detection.



THE CLOUD CHAMBER

A unique device for detection and measurement is the Cloud Chamber, invented by the British physicist Charles Wilson in 1911. The Cloud chamber makes it possible to

visually see the path of ionizing radiation thus making it possible to photograph it. The cloud chamber consists of a plastic or glass container, which sits on dry ice. A dark cloth is saturated with alcohol and placed around the inside of the container near the top. A small radioactive material may be suspended from the lid of the container. In the chamber, the alcohol evaporates from the cloth and condenses as it reaches the cold region created by the dry ice at the floor of the container. Just above the floor of the chamber there is a region where the alcohol vapor does not condense unless there are seeds around, so that drops of alcohol can form. This condition

